

EMERGING MATERIALS, RARE EARTHS AND CRITICAL MINERALS

PROPERTY STRUCTURE PROCESSING MAP → ADVANCED MATERIALS FAMILY → REE CLASSIFICATION RAIL → CATEGORY FIREWALL → PERMANENT MAGNET CAPABILITY LADDER → BATTERY MINERAL ROLE MATRIX

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g2 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 PROPERTY STRUCTURE PROCESSING MAP

PROPERTY STRUCTURE PROCESSING MAP COMPOSITION + BONDING INTRINSIC BEHAVIOUR MOLECULAR STRUCTURE PROPERTY PATHWAYS MICROSTRUCTURE + DEFECTS LOCAL PERFORMANCE CHANGES STRUCTURE AND DEFECTS

COMPOSITION + BONDING → intrinsic behaviour

CRYSTAL / MOLECULAR STRUCTURE → property pathways

MICROSTRUCTURE + DEFECTS → local performance

PROCESSING → changes structure and defects

STARTING CONDITION / INPUT

- COMPOSITION + BONDING → intrinsic behaviour
- CRYSTAL / MOLECULAR STRUCTURE → property pathways

SCIENTIFIC OR ENGINEERING MECHANISM

- MICROSTRUCTURE + DEFECTS → local performance
- PROCESSING → changes structure and defects

OUTPUT / FAILURE BOUNDARY

- TEST → strength != stiffness != toughness != hardness
- COMPOSITION + BONDING → intrinsic behaviour

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TEST → strength != stiffness != toughness != hardness.

01

STAGE 01 ADVANCED MATERIALS FAMILY

ADVANCED MATERIALS FAMILY ADVANCED MATERIAL ENGINEERED DEMANDING FUNCTION MATRIX + REINFORCEMENT CARBON-FIBRE COMPOSITE HIGH SPECIFIC STRENGTH/STIFFNESS SMART MATERIAL STIMULUS-RESPONSIVE PROPERTY

ADVANCED MATERIAL → engineered demanding function

COMPOSITE → matrix + reinforcement

CARBON-FIBRE COMPOSITE → high specific strength/stiffness

SMART MATERIAL → stimulus-responsive property

CONCEPT / ARCHITECTURE

- ADVANCED MATERIAL → engineered demanding function
- COMPOSITE → matrix + reinforcement

MECHANISM / APPLICATION

- CARBON-FIBRE COMPOSITE → high specific strength/stiffness
- SMART MATERIAL → stimulus-responsive property

READINESS / STATUS LIMIT

- EXAMPLES → shape-memory alloy
- piezoelectric material

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SMART MATERIAL → stimulus-responsive property; EXAMPLES → shape-memory alloy | piezoelectric material.

02

STAGE 02 REE CLASSIFICATION RAIL

REE CLASSIFICATION RAIL 15 LANTHANIDES + SCANDIUM + YTTRIUM HEAVY OWNER EXAMPLES CHIEFLY LIGHT-REE-RICH INDIAN CONTEXT GEOLOGICAL OCCURRENCE SEPARATED MAGNET INPUT

LIGHT → La

Ce

Pr

Nd

HEAVY OWNER EXAMPLES → Dy

Tb

Y

SYSTEM / DEVICE / INSTITUTION

- REE = 15 LANTHANIDES + SCANDIUM + YTTRIUM
- LIGHT → La

DECISIVE TECHNICAL DISTINCTION

- HEAVY OWNER EXAMPLES → Dy
- MONAZITE → chiefly light-REE-rich Indian context

STATUS / UNIT / EVIDENCE LIMIT

- TRAP → geological occurrence != separated magnet input
- REE = 15 LANTHANIDES + SCANDIUM + YTTRIUM

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → geological occurrence != separated magnet input.

03

STAGE 03 CATEGORY FIREWALL

CATEGORY FIREWALL CHEMICAL GROUP RARE EARTH ELEMENTS POLICY CATEGORY CRITICAL MINERALS STATUTORY ROUTE PART D EXPLORATION ROUTE

SYSTEM / DEVICE / INSTITUTION

- CHEMICAL GROUP → rare earth elements
- POLICY CATEGORY → critical minerals

DECISIVE TECHNICAL DISTINCTION

- STATUTORY ROUTE → Part D
- EXPLORATION ROUTE → Growth Schedule

STATUS / UNIT / EVIDENCE LIMIT

- LITHIUM / COBALT / NICKEL / GRAPHITE → critical may be, REE no
- CHEMICAL GROUP → rare earth elements

03

SYSTEM / DEVICE / INSTITUTION

- REE = 15 LANTHANIDES + SCANDIUM + YTTRIUM
- LIGHT → La

DECISIVE TECHNICAL DISTINCTION

- HEAVY OWNER EXAMPLES → Dy
- MONAZITE → chiefly light-REE-rich Indian context

STATUS / UNIT / EVIDENCE LIMIT

- TRAP → geological occurrence != separated magnet input
- REE = 15 LANTHANIDES + SCANDIUM + YTTRIUM

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → geological occurrence != separated magnet input.

04

STAGE 03

CATEGORY FIREWALL

CATEGORY FIREWALL

CHEMICAL GROUP

RARE EARTH ELEMENTS

POLICY CATEGORY

CRITICAL MINERALS

STATUTORY ROUTE

PART D

EXPLORATION ROUTE

SYSTEM / DEVICE / INSTITUTION

- CHEMICAL GROUP → rare earth elements
- POLICY CATEGORY → critical minerals

DECISIVE TECHNICAL DISTINCTION

- STATUTORY ROUTE → Part D
- EXPLORATION ROUTE → Seventh Schedule

STATUS / UNIT / EVIDENCE LIMIT

- LITHIUM / COBALT / NICKEL / GRAPHITE → critical may be, REE no
- CHEMICAL GROUP → rare earth elements

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LITHIUM / COBALT / NICKEL / GRAPHITE → critical may be, REE no.

05

STAGE 04

PERMANENT MAGNET CAPABILITY LADDER

PERMANENT MAGNET CAPABILITY LADDER

MAGNETIC PERFORMANCE

HIGH-TEMPERATURE QUALIFICATION

ORE → SEPARATION → OXIDE / METAL

METAL → ALLOY → MAGNET

STARTING CONDITION / INPUT

- ORE → SEPARATION → OXIDE / METAL
- METAL → ALLOY → MAGNET

MAGNET → MOTOR / ELECTRONICS / DEFENCE

SCIENTIFIC OR ENGINEERING MECHANISM

- MAGNET → MOTOR / ELECTRONICS / DEFENCE
- Nd → magnetic performance

Nd → magnetic performance

OUTPUT / FAILURE BOUNDARY

- Dy / Tb → high-temperature qualification
- ORE → SEPARATION → OXIDE / METAL

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Nd → magnetic performance; Dy / Tb → high-temperature qualification.

06

STAGE 05

BATTERY MINERAL ROLE MATRIX

BATTERY MINERAL ROLE MATRIX

CHARGE-CARRIER SYSTEM INPUT

NICKEL + COBALT

NCA CATHODES

COMMON LITHIUM-ION ANODE

NEITHER NICKEL NOR COBALT

AVOIDS LITHIUM

SYSTEM / DEVICE / INSTITUTION

LITHIUM → charge-carrier system input

NICKEL + COBALT → NMC / NCA cathodes

DECISIVE TECHNICAL DISTINCTION

GRAPHITE → common lithium-ion anode

LFP → neither nickel nor cobalt

STATUS / UNIT / EVIDENCE LIMIT

SODIUM-ION → avoids lithium

LITHIUM → charge-carrier system input

SYSTEM / DEVICE / INSTITUTION

- LITHIUM → charge-carrier system input
- NICKEL + COBALT → NMC / NCA cathodes

DECISIVE TECHNICAL DISTINCTION

- GRAPHITE → common lithium-ion anode
- LFP → neither nickel nor cobalt

STATUS / UNIT / EVIDENCE LIMIT

- SODIUM-ION → avoids lithium
- LITHIUM → charge-carrier system input

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NICKEL + COBALT → NMC / NCA cathodes.

STAGE 06

NAMED APPLICATION PORTFOLIO

NAMED APPLICATION PORTFOLIO

HIGH-TECH ELECTRONICS

ELECTRICAL SYSTEMS

SPECIALISED OPTICS

TITANIUM-BEARING MINERALS

SEPARATE POLICY

GEOGRAPHY INPUT

RARE-EARTH VALUES + THORIUM-BEARING CONTEXT

COPPER → electrical systems

CONCEPT / ARCHITECTURE

- GALLIUM / GERMANIUM → high-tech electronics
- COPPER → electrical systems

REE → magnets

MECHANISM / APPLICATION

- REE → magnets
- specialised optics

specialised optics

READINESS / STATUS LIMIT

- TITANIUM-BEARING MINERALS → separate policy / geography input
- MONAZITE → rare-earth values + thorium-bearing context

EMERGING MATERIALS, RARE EARTHS AND CRITICAL MINERALS • TILE 2/4 • same master rows 2498-5732 of 10728 • continues below

06

STAGE 06 NAMED APPLICATION PORTFOLIO

NAMED APPLICATION PORTFOLIO

HIGH-TECH ELECTRONICS

ELECTRICAL SYSTEMS

SPECIALISED OPTICS

TITANIUM-BEARING MINERALS

SEPARATE POLICY

GEOGRAPHY INPUT

RARE-EARTH VALUES + THORIUM-BEARING CONTEXT

GALLIUM / GERMANIUM → high-tech electronics

COPPER → electrical systems

REE → magnets

specialised optics

CONCEPT / ARCHITECTURE

- GALLIUM / GERMANIUM → high-tech electronics
- COPPER → electrical systems

MECHANISM / APPLICATION

- REE → magnets
- specialised optics

READINESS / STATUS LIMIT

- TITANIUM-BEARING MINERALS → separate policy / geography input
- MONAZITE → rare-earth values + thorium-bearing context

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TITANIUM-BEARING MINERALS → separate policy / geography input.

07

STAGE 07 MINE TO TECHNOLOGY VALUE CHAIN

MINE TO TECHNOLOGY VALUE CHAIN

RECOVERED FEED WITH PURITY/ECONOMIC LIMITS

EXPLORE → ASSESS → DEVELOP → EXTRACT

BENEFICIATE → SEPARATE / REFINE

EXPLORE → ASSESS → DEVELOP → EXTRACT

BENEFICIATE → SEPARATE / REFINE

OXIDE / METAL / ALLOY → COMPONENT

COMPONENT → BATTERY / MAGNET / CHIP / SYSTEM

RECYCLE → recovered feed with purity/economic limits

STARTING CONDITION / INPUT

- EXPLORE → ASSESS → DEVELOP → EXTRACT
- BENEFICIATE → SEPARATE / REFINE

SCIENTIFIC OR ENGINEERING MECHANISM

- OXIDE / METAL / ALLOY → COMPONENT
- COMPONENT → BATTERY / MAGNET / CHIP / SYSTEM

OUTPUT / FAILURE BOUNDARY

- RECYCLE → recovered feed with purity/economic limits
- EXPLORE → ASSESS → DEVELOP → EXTRACT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: COMPONENT → BATTERY / MAGNET / CHIP / SYSTEM.

08

STAGE 08 DEPENDENCY HEAT MAP

DEPENDENCY HEAT MAP

SECURE MINE ALONE IS PARTIAL SECURITY

UPSTREAM → GEOLOGY

MIDSTREAM → SEPARATION

UPSTREAM → geology

MIDSTREAM → separation

DOWNSTREAM → cell

ENABLERS → technology

CONCEPT / ARCHITECTURE

- UPSTREAM → geology
- MIDSTREAM → separation

MECHANISM / APPLICATION

- DOWNSTREAM → cell
- ENABLERS → technology

READINESS / STATUS LIMIT

- VERDICT → secure mine alone is partial security
- UPSTREAM → geology

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → secure mine alone is partial security.

09

STAGE 09 CIRCULAR MATERIALS LOOP

CIRCULAR MATERIALS LOOP

RE-ENTER MANUFACTURE

SUBSTITUTE + EFFICIENCY

REDUCE PRIMARY-MATERIAL PRESSURE

MUST REMEMBER

CRITICAL-MINERAL ANALYSIS FOLLOWS GEOLOGY, RESOURCE/RESERVE, MINING

DESIGN → durability

USE → reuse / remanufacture

COLLECT → sort → pre-process

RECOVER → purify → re-enter manufacture

SUBSTITUTE + EFFICIENCY → reduce primary-material pressure

CONCEPT / ARCHITECTURE

- DESIGN → durability
- USE → reuse / remanufacture

MECHANISM / APPLICATION

- COLLECT → sort → pre-process
- RECOVER → purify → re-enter manufacture

READINESS / STATUS LIMIT

- SUBSTITUTE + EFFICIENCY → reduce primary-material pressure
- MUST REMEMBER: Critical-mineral analysis follows geology, resource/reserve, mining,....

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Critical-mineral analysis follows geology, resource/reserve, mining,....

10

STAGE 10 MINE LIFE SAFEGUARD CHAIN

09

STAGE 09 CIRCULAR MATERIALS LOOP

CIRCULAR MATERIALS LOOP RE-ENTER MANUFACTURE SUBSTITUTE + EFFICIENCY REDUCE PRIMARY-MATERIAL PRESSURE MUST REMEMBER CRITICAL-MINERAL ANALYSIS FOLLOWS GEOLOGY, RESOURCE/RESERVE, MINING



CONCEPT / ARCHITECTURE

- DESIGN → durability
- USE → reuse / remanufacture

MECHANISM / APPLICATION

- COLLECT → sort → pre-process
- RECOVER → purify → re-enter manufacture

READINESS / STATUS LIMIT

- SUBSTITUTE + EFFICIENCY → reduce primary-material pressure
- MUST REMEMBER: Critical-mineral analysis follows geology, resource/reserve, mining,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Critical-mineral analysis follows geology, resource/reserve, mining,.

10

STAGE 10 MINE-LIFE SAFEGUARD CHAIN

MINE-LIFE SAFEGUARD CHAIN CUMULATIVE IMPACT + CONSULTATION TAILINGS CONTROLS FRA DUE DILIGENCE STABLE DUMPS NATIVE RESTORATION FINANCE + MONITORING; PLAN ACHIEVED OUTCOME

OPERATIONS → dust	noise	water	tailings controls
COMMUNITY → FRA due diligence	disclosure	grievance	
RECLAMATION → topsoil	stable dumps	native restoration	

STARTING CONDITION / INPUT

- SITING → cumulative impact + consultation
- OPERATIONS → dust
- tailings controls

SCIENTIFIC OR ENGINEERING MECHANISM

- COMMUNITY → FRA due diligence
- RECLAMATION → topsoil
- stable dumps

OUTPUT / FAILURE BOUNDARY

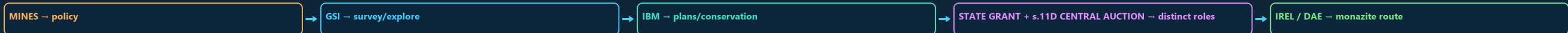
- native restoration
- CLOSURE → finance + monitoring; plan != achieved outcome
- CLOSE DISTINCTION: Critical is not necessarily rare, rare earths are not all scarce,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Critical is not necessarily rare, rare earths are not all scarce,.

11

STAGE 11 INDIA INSTITUTIONS STATUS AND PYQ RAIL

INDIA INSTITUTIONS STATUS AND PYQ RAIL STATE GRANT + S.11D CENTRAL AUCTION DISTINCT ROLES MONAZITE ROUTE OVERSEAS ROUTE DEEP-OCEAN EXPLORATION BOUNDARY EVIDENCE LIMIT NAME MINERAL/MATERIAL, VALUE-CHAIN NODE



CONCEPT / ARCHITECTURE

- MINES → policy
- GSI → survey/explore
- IBM → plans/conservation

MECHANISM / APPLICATION

- STATE GRANT + s.11D CENTRAL AUCTION → distinct roles
- IREL / DAE → monazite route
- KABIL → overseas route

READINESS / STATUS LIMIT

- MoES / NIOT / ISA → deep-ocean exploration boundary
- REPORT → LAW → APPROVAL → MOU → EXPLORATION → PRODUCTION → OUTCOME
- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Name mineral/material, value-chain node,...

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Name mineral/material, value-chain node,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Name mineral/material, value-chain node,.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT THE KEY ADVANCED DISTINCTION IS BETWEEN GEOLOGICAL PRESENCE AND A COUNTRY MAY HAVE RESOURCES BUT STILL REMAIN VULNERABLE THEREFORE, CRITICAL-MINERAL POLICY IS SIMULTANEOUSLY A MINING POLICY, INDUSTRIAL MINISTRY OF MINES
POLICY AND MISSION-LEVEL ANCHOR. THE NATIONAL GEOSCIENTIFIC SURVEY AND EXPLORATION AGENCY — NOT MINING-PAN APPROVAL, SCIENTIFIC MINING AND MINERAL CONSERVATION

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH

- CAUTION: The key advanced distinction is between geological presence and usable strategic capability .
- CAUTION: A country may have resources but still remain vulnerable if refining and component manufacturing are concentrated elsewhere.
- CAUTION: Therefore, critical-mineral policy is simultaneously a mining policy, industrial policy, trade policy and strategic policy.
- Ministry of Mines: policy and mission-level anchor.

READINESS, UNIT OR STATUS LIMIT

- GSI: the national geoscientific survey and exploration agency — not the mine developer, lease-granting authority or sectoral regulator.
- IBM: mining-plan approval, scientific mining and mineral conservation
- State governments grant concessions, with the Centre auctioning Part D minerals under s.11D .
- IREL (India) Ltd (DAE): monazite processing for rare earths and thorium.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE

- KABIL: overseas sourcing/exploration instrument.
- MoES / NIOT — Deep Ocean Mission: seabed polymetallic-nodule exploration under an ISA contract.
- Mineral Security Partnership: plurilateral investment-coordination platform India has joined.
- NCMM: mission framework covering exploration, mining, processing, recycling and stockpiles.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.